

## 1. Question

Write a Java program to demonstrate exception handling using `IllegalArgumentException` when an invalid thread priority is assigned.

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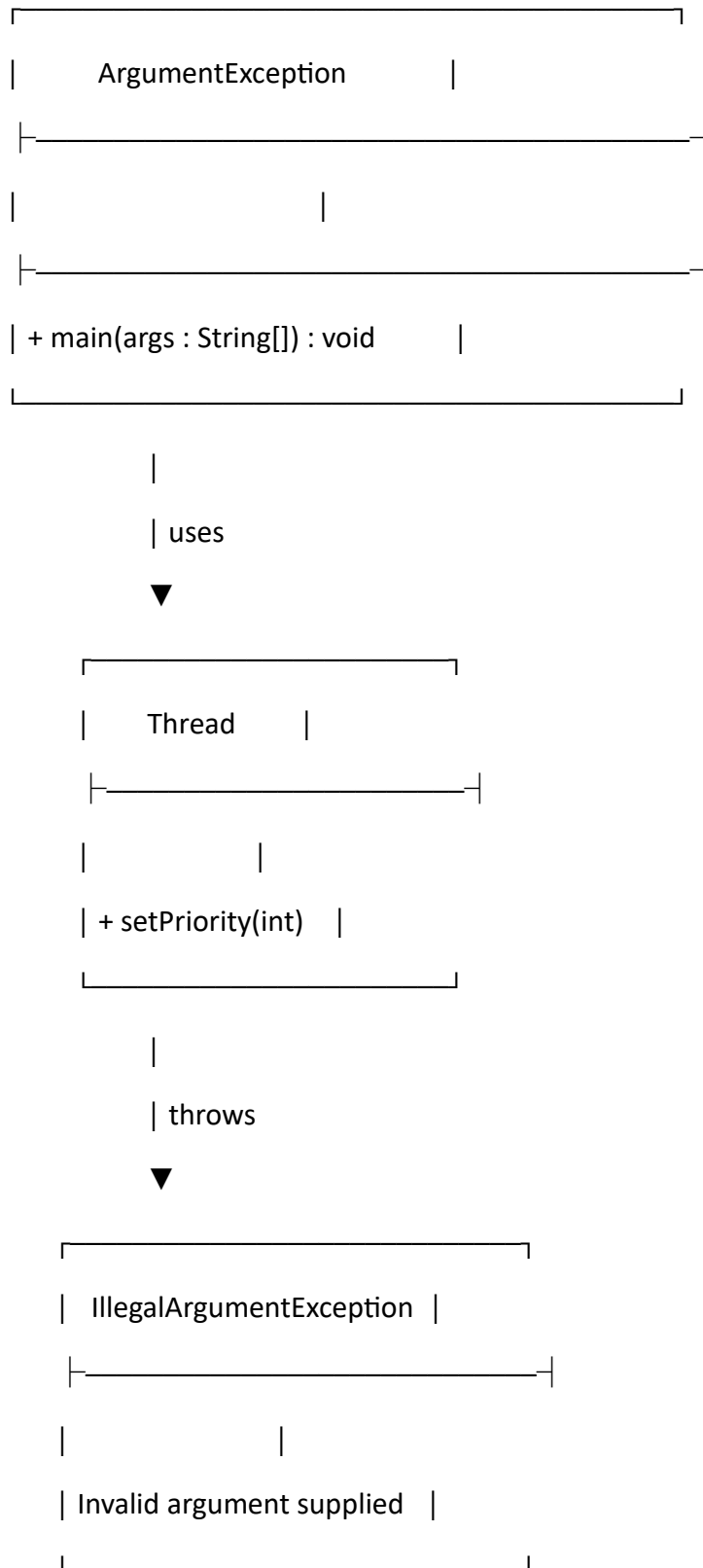
## 2. Steps for Execution

1. Open a Java IDE or command prompt.
2. Create a package named `javapractice`.
3. Create a class named `ArgumentException`.
4. Create a `Thread` object.
5. Use a try block to set the thread priority.
6. Set the priority to 15.
7. Since the valid thread priority range is **1 to 10**, Java throws `IllegalArgumentException`.
8. The catch block catches the exception.
9. Display "Program ended.".
10. Compile and execute the program.

## 3. Algorithm

1. **Start**
  2. Create a `Thread` object named `thread`.
  3. Begin the try block.
  4. Display "Setting thread priority".
  5. Set the thread priority to 15.
  6. Check the priority value.
  7. Since 15 is outside the valid range of **1–10**, an `IllegalArgumentException` occurs.
  8. Catch the exception using the catch block.
  9. Display "Program ended.".
  10. **Stop**
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#### 4. UML Class Diagram



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#### 5. Code

```
package javapractice;

public class ArgumentException {
    public static void main(String[] args) {

        Thread thread = new Thread();

        try {
            System.out.println("Setting thread priority");

            thread.setPriority(15);

            System.out.println("Priority set successfully.");
        }
        catch (IllegalArgumentException e) {
            System.out.println("Program ended.");
        }
    }
}
```

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## 6. Output

Setting thread priority

Program ended.

The statement:

```
System.out.println("Priority set successfully.");
```

is **not executed** because the exception occurs at `setPriority(15)`.

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## 7. Result

**Thus, the Java program successfully demonstrates exception handling using `IllegalArgumentException` when an invalid thread priority is assigned.**